

DIPLOMACY AND THE SEARCH FOR EXTRATERRESTRIAL INTELLIGENCE (SETI)

ALLEN E. GOODMAN

School of Foreign Service, Georgetown University, Washington, DC 20057 USA

Abstract—Recent advances in astronomy indicate that the universe is spectacularly vast. Apparently, our Sun is like billions of other stars around which planets may circle; within just a thousand light years of Earth, for example, there are more than a million stars like the Sun. Within the Milky Way (only one of 100 billion observable galaxies) are an estimated 300-400 billion stars. Scientists now think that there are probably between 10^{21} and 10^{22} stars in the universe.

With so many stars, it is highly probable that we are not alone.

Just how likely extraterrestrial intelligence is or in how many places and varieties it will be found is impossible now to say. Serious scientists have constructed an equation to calculate the possibility, but the range of estimates of the number of other civilizations in the universe is so wide - anywhere from 200 million to less than 10 - and our understanding of astrophysics and cosmic evolution so primitive, that we could encounter extraterrestrials (ETs) well before it can be proved on paper that they have existed or could exist[1].

In fact, the capability to search for extraterrestrial intelligence will increase over the next five years at a rate and scale greater than at any time in human history. For example, a Harvard University observatory announced in October 1985 that it had linked a radio telescope to a new computerized multichannel spectrum analyzer and created a listening post (known as Project Meta) capable of simultaneously scanning up to 8.4 million radio channels. This development replaces a system whose capacity to search simultaneously and record radio signals was a mere 180,000 radio channels. The Hubble Space Telescope, when it is operational, will extend our present vision some seven times and possibly to the edge of the universe itself. If other planetary systems exist, this new instrument might locate them. And by the end of this decade, NASA will probably have begun a search for signals across the entire microwave spectrum while other researchers will have developed the instruments to search for large planets in orbit around some 200 nearby stars. If there is someone or thing out there trying to communicate with Earth, within the next three to five years we increasingly will know where to look and listen.

This essay focuses on what to do when these or other research projects indicate that extraterrestrial intelligence exists.

1. THE NEED FOR DIPLOMACY

While the body of literature on the science of the search for extraterrestrial intelligence (SETI) is serious and growing[2], there are no studies of the diplomacy for handling the first contact. In part, this is due to skepticism about the feasibility of interstellar travel[3].

The predominant expectation in scientific circles is that such contact will be via electromagnetic radiation at microwave frequencies[4] (where natural cosmic status is at a minimum). Scientific measure disabuses me of the expectation of facile physical contact: light years of distance separates the Earth from those stars nearest it which might be capable of supporting intelligent life. The Pioneer and Voyager satellites, travelling at a speed of 25,000 miles per hour, will require more than a quarter million years to reach such stars[5].

Radio communication, travelling at the speed of light[6], would certainly be more swift and practicable. The round-trip transmission time for a message and response still, of course, could be an entire lifetime; it could also be a quarter of a century or less, if some of the Sun-like stars nearest to Earth were the origin of signals from extraterrestrial intelligence. Recent data from the Infrared Astronomy Satellite,

moreover, suggest that about 10% of these nearby stars may be surrounded by planets in some stage of formation[7].

On none of these issues am I in a position, as a political scientist, to take a stand. This article is based on the assumption that contact with extraterrestrial intelligence will be made in some way and that it is prudent to begin thinking about how we would and should respond.

2. EXISTING TREATIES

Unlike other aspects of space exploration SETI is not the subject of any international treaty, agreement, or national research policy. The 1967 UN Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, which has been ratified by 86 countries, does establish some general principles that by extrapolation could be applied to SETI programs. The Treaty provides that the exploration of space should be undertaken for the common benefit of mankind, establishes that states are responsible for consultations about activities, public or private, that its nationals undertake in space, requires international consultations about activities that might interfere with those of other states, and accords

to astronauts the status of envoy. But this treaty does not go far enough in my view to cover the contingency of success in SETI, especially the conditions under which such a development should be announced and the information derived from it shared.

The 1979 UN-sponsored Agreement governing the Activities of States on the Moon and Other Celestial Bodies is the other international document that is at all relevant. This agreement suggests that there should be prompt and public release of information when states discover any indication of organic life in space. But it has been signed by only 11 countries and ratified by five: none of these countries are presently engaging in SETI or planning to do so. And even if this agreement were adopted widely, I do not think it would be effective in covering the range of ways SETI is conducted or the possibility that radio astronomers might be uncertain for many years that the signals they detect are indicative of organic life[8].

3. CURRENT SETI PLANS

With respect to national space exploration and research policies, I have surveyed all current and past SETI projects and their directors and so far have found that no technical cooperation agreements have been developed to cover a successful SETI[9]. There is, in fact, considerable variation in opinion on what should be done, were a search to prove successful. Such opinion ranges from the view that there should be prompt and public announcement, to the need to process any announcement through national scientific and security agencies, to the view that what to say and when should be decided by the head of state in the country where the discovery is made.

The governments, agencies and observatories currently engaged in SETI differ widely about how to respond to a successful contact. There are now more than two dozen ongoing searches devoted entirely to SETI; 8 are in the US; 5 in the USSR and 2 others in Europe. Some are government efforts undertaken by civilian space agencies, some are experiments by or on military installations; and some are under entirely private research auspices[10].

The private US SETI searchers have not yet thought through the protocols they would follow in the event of success. NASA's SETI Project has not yet established an official protocol although one is currently being developed. Dr. Bernard Oliver, the Project Director, told me

that his first step would be to seek confirmation of a signal detection by checking with other observatories and thus making the fact of potential discovery more or less public. Soviet SETI searchers, in contrast, do not apparently practice such a confirmation procedure and already have announced several contacts, only to have discovered later that their receivers picked up secret, encoded US intelligence satellite signals. The experience of British radio astronomers provides yet another variant on how such a discovery might be handled. When (in 1967) they discovered what was first thought to be signals from interstellar beacons and later proved to be pulsars, they refrained from any announcement for more than half a year while they tried to develop an explanation.

4. SUGGESTED INTERNATIONAL CODE OR CONDUCT

Given these examples, there is a need for discussion that anticipates, at least, successful electromagnetic contact from another civilization in space. The significance of such an event surely should compel all countries to recognize their common heritage and tenancy on Earth within the much vaster universe. But will the requirement for global diplomacy be overwhelmingly self-evident to all nations?

For example, we should be prepared at least for the possibility that we will intercept a communication signal not meant for us but one which contains a wealth of data about the source which sent it. This type of "communication" should be treated as global property, free to all. But, at present, there is no guarantee that a nation which intercepts it would feel obliged to share. In fact, I suggest that the temptation to restrict dissemination - especially if the contents of the signal contain important scientific and technological data that would be commercially or militarily valuable - would be quite strong if such a signal were intercepted by a superpower or during a time of global or regional tension. There is a real risk, moreover, that SETI could become politicized as the military investment in national space exploration activities increases - as, for example, it is projected to do in the United States in the wake of the Challenger disaster - or if space-based research facilities play a major role in developing strategic defense systems.

One way to avoid this risk is to develop now an international code of conduct based on the principle that the search for and any contact with extraterrestrials is to be undertaken on

behalf of the Earth and for the purpose of sharing and extending knowledge. The following operational principles derive from this basic tenet:

- Scientists, observatories, and governmental agencies should freely and publicly announce the detection of any communication or radio signal suspected to originate from a non-natural extraterrestrial source. Discovery of an extraterrestrial spacecraft or habitat should also be freely and publicly announced.
- Any response to extraterrestrial communication or signals or plan for investigation of extraterrestrial artifacts should be formulated by a process of international consultation among concerned governmental authorities and scientists.
- Visiting extraterrestrials should be regarded as envoys entitled to diplomatic immunity, protection, or aid in the event of an accident, distress or emergency. Such status should be accorded ET's regardless of the form in which they arrive or their apparent communication capability.
- In the event that extraterrestrials appear to pose a threat to human health or peace, no nation should act without first consulting the Security Council of the United Nations.

This code of conduct could be developed and negotiated under the auspices of the United Nations, but doing so runs the risk that it would become politicized along east-west or north-south lines in a way that would make it impossible for many key countries to ratify. An alternative process, suggested by the Deputy Chief of the Outer Space Affairs Division of the UN, would be to begin by developing a technical cooperation agreement between those agencies and scientists engaged in SETI and follow this up with governmental agreements only when tenable and widely accepted protocols have emerged. Such an agreement could then become the basis for an international treaty negotiated between the concerned countries; as such, the agreement could be opened to general ratification through the United Nations.

While my survey of international law and SETI projects found that such a code of conduct does not exist, it also revealed growing concern by those engaged in exploring outer space about the impact of their work on national space policy generally, and whether their research would be politicized, if successful. Such concern has been discussed by only a handful of diplomats and national leaders[11]; those who presently set the foreign policy agenda do not regard SETI as a

pressing issue. Whatever one may think about the likelihood of extraterrestrial civilizations or how well earthlings might manage contact with them, however, development now of an international code of conduct could assure that the present and projected SETI activities of all countries take place in an atmosphere of international cooperation and public trust.

5. CONCLUSIONS

There are two principle objections to the effort to develop a SETI code of conduct in advance of a successful contact. The first is that countries or scientists will not necessarily feel bound by such a code once such a contact is made, especially if it happens in an unanticipated way or form. This is a standard - and discredited - line of argument against most international laws and treaties. Of course there are scofflaws and some international treaties are not consistently honored. But an overwhelming number of laws and treaties are observed and, generally speaking, nations find it in their interest to honor such conventions.

A SETI code of conduct in advance of a contact will not assure that nations will honor it. However, it will cause those contemplating proceeding or tempted to proceed in a different way to think twice about ignoring internationally endorsed rules.

The second general type of objection to a SETI code of conduct is that contact may be made by facilities or space explorations or missions where secrecy is required or expected. There is no question that space exploration and research is increasingly a mission of national military establishments and will become more so as the search for an effective, space-based Strategic Defense Initiative intensifies. Should these trends continue, there is all the more reason to try to assure that a discovery of significance to an entire planet - and which may require the brains of many nations to appreciate and interpret - should be shielded from the pressures of secrecy and national rivalries. A code of conduct can provide such a shield.

So the initiative not only of searching for extraterrestrial intelligence but also of anticipating its diplomatic consequences lies with the scientific community today. This should by now be a familiar role for scientists and explorers. For throughout human history the discoverers of new worlds are usually the ones who, at least initially, decide how to relate to the civilizations they encounter.

REFERENCES

1. See for example the layman's discussion in R. T. Rood and J. S. Trefil, *Are We Alone?* Charles Scribner's Sons, New York (1983). The quoted range of estimates is from p. 7. M. Harwit's *Cosmic Discovery: The Search, Scope and Heritage of Astronomy*. Basic Books, New York (1981) suggests that with respect to our knowledge of cosmic phenomena, mankind has yet to discover two-thirds of what exists.
2. For an excellent, annotated bibliography, see C. L. Seeger's "Selected SETI References and Reading List" in F. Drake, J. H. Wolfe and C. L. Seeger, eds. *SETI Science Working Group Report*, NASA Technical Paper 2244, Washington, DC, pp. 101-110 (1983). See also E. F. Mallove, et al., *A Bibliography on the Search for Extraterrestrial Intelligence*, NASA Conference Publication 1021, Washington, DC (1978).
3. More than one point of view exists in the discussion of interstellar flight. Two scientists who dismiss the practicality of interstellar flight are J. H. Wolfe, "On the Question of Interstellar Travel" in a paper prepared for International Astronomical Union Symposium 112 (1984) and B. M. Oliver in "Interstellar Relativistic Rocketry" (March 1983). Dr. Oliver is chief of NASA's SETI Program office, and has concluded that:

"The energy and time required for interstellar travel cast doubt on its existence even for very advanced societies. The cost and time scale of interstellar commerce seem prohibitive. Without commerce with the colonies, the motivation for colonization is not evident. Certainly it offers no solution to population pressures.
 "Even if intelligent life is common in the universe it should come as no surprise that we have not been visited. Perhaps that life is intelligent enough to realize how much cheaper and easier interstellar communication is, and how nearly equally rewarding. Perhaps interstellar travel and colonization do not exist because interstellar communication does (page 14)."
- Support for interstellar travel, originating in the *Daedalus* study of 1978, has appeared as a frequent theme in the *Journal of the British Interplanetary Society* and elsewhere. Forward, Mallove, and Paprotny have compiled a bibliography on interstellar flight and communication which elucidates the diversity of opinion on this issue.
4. i.e., one to ten billion Hertz. Within this range, present and projected SETI searches are focused on the roughly 300 million Hertz between the characteristic radio wave emissions of the neutral hydrogen atom and hydroxyl molecules. The scanning of these frequency bands started in 1960. Then the capacity to listen was limited to one channel at a time. In addition, "existing terrestrial microwave radio telescope technology is sufficiently sensitive to detect reasonable signals...from distances of 1000 ly (light years) or more. Within that distance are more than a million sun-like stars..." from "Fundamental Concepts of Microwave Searches for Extraterrestrial Intelligence" in Drake, et al., NASA TP 2244, p. 1.
5. Aboard each of the two voyager spacecraft, launched by the United States on August 20 and September 5, 1977, is included a gold-plated phonograph record that can produce both sound and pictures, and the equipment to play it. The recorded message includes greetings in 55 of the world's languages; sounds, both natural and human, and a representative sample of the world's music. Pioneer 10, launched on March 3, 1972, carries a plaque depicting the human male and female, a representation of the solar system, and accompanying symbols from which can be interpreted the origin of the spacecraft in space and time.
6. For the moment - and the foreseeable future - any communication is bound by the speed of light. But this implies such a limit on our capability to explore the universe and to send and receive radio signals throughout it that I don't think we should indefinitely assume that this barrier will remain fixed. Indeed, there are already some tantalizing hints that radio signals emitted by Quasars may travel faster than the speed of light. See G. B. Field and E. J. Chaisson, *The Invisible Universe*. Birkhauser, Boston, p. 110 (1985), and R. Breuer, *Contact with the Stars*. W. H.

- Freeman & Co., Oxford and San Francisco, pp. 196-198 (1982).
7. J. S. Davidson, E. T. come home: communicating with the extraterrestrial. *Language and Society* 13, (Spring 1984). M. M. Waldrop, First Sightings. *Science* 85, 26-33 (June 1985).
 8. The Second UN Conference on the Exploration and Peaceful Uses of Outer Space in 1982 has suggested in one of its many recommendations (#310) that "SETI...should be considered a joint international effort..." This recommendation, of course, has no real standing in international law and thus far has not been taken up even by the UN General Assembly.
 9. What effect such a lacuna might have on how the first contact is handled and intercepted is portrayed in Dr. C. Sagan's new novel *Contact*. Simon and Schuster, New York (1985).
 10. See Jill Tarter, Summary of SETI Observing Programs (November 1985). The US government SETI program is described in "NASA's SETI Program," a February 1985 Fact Sheet. A similar fact sheet on the Soviet progress is not available although Soviet scientists have had a long-standing interest in SETI and an apparently well funded research and observation program.
 11. See, for example, M. A. G. Michaud, Interstellar Negotiation. *Foreign Service Journal* (December 1972).